

DATA ANALYSIS REPORT ON ROAD ACCIDENTS IN THE UK: Insights for Coretech Labs

Presented by: Pod POLARIS
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INTRODUCTION

❖ CoreTech Labs stands at the forefront of modern technology:

- ✓ Software development
- ✓ AI integration
- ✓ data-driven innovation.

❖ Collaboration with 10Alytics:

- ✓ Response to rising road safety concerns.

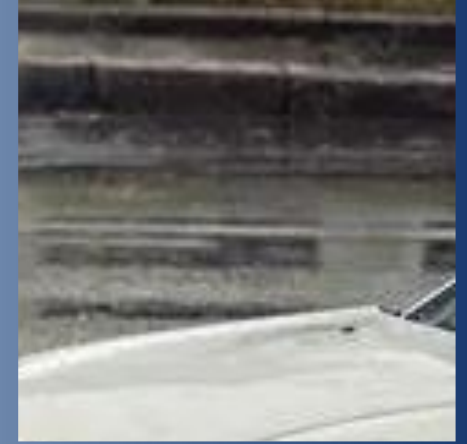
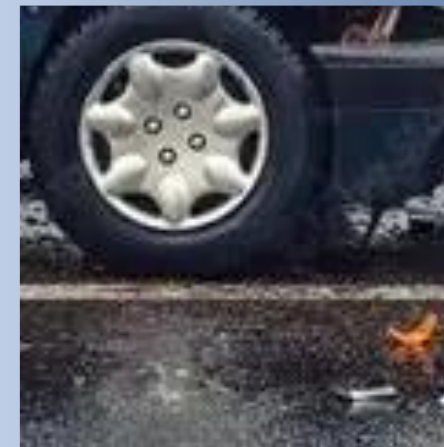
❖ This presentation will explore UK accident trends:

- ✓ From 2021 to 2022
- ✓ Aim to uncover insights
- ✓ Inform effective safety strategies.



PROBLEM STATEMENT

- ❖ Road accidents remain a major public safety concern.
- ❖ Accident severity is influenced by environmental conditions and regulatory factors.
- ❖ Available accident data is largely unstructured and difficult to interpret.
- ❖ Stakeholders face challenges in extracting actionable insights from raw datasets.
- ❖ Dashboards and data visualization tools are underutilized in decision-making processes.



OBJECTIVES

- ❖ To Clean and Standardize the Accident Dataset
- ❖ To Analyze the Frequency and Severity of Accidents
- ❖ To Identify High-Risk Factors Contributing to Accidents
- ❖ To Perform Descriptive Statistical Analysis
- ❖ To Develop an Interactive Excel Dashboard for Data Visualization
- ❖ To Provide Data-Driven Recommendations for Road Safety Improvements



DATA DESCRIPTION

❖ The dataset consists of UK road traffic accident records from 2021 to 2022

❖ Dataset contains over 300k rows of both numerical and categorical variables.

❖ Key variables include accident time, location, speed limit, accident severity, road characteristics, environmental conditions, and vehicle involvement.

❖ Tool used was excel

Accident Index	Accident Date	Day of Week	Accident Severity	Latitude	Light Conditions
200901BS70001	01/01/2021	Thursday	Serious	51.512273	Daylight
200901BS70002	05/01/2021	Monday	Serious	51.514399	Daylight
200901BS70003	04/01/2021	Sunday	Slight	51.486668	Daylight
200901BS70004	05/01/2021	Monday	Serious	51.507804	Daylight
200901BS70005	06/01/2021	Tuesday	Serious	51.482076	Darkness - lights lit
200901BS70006	01/01/2021	Thursday	Slight	51.493415	Daylight

Longitude	Number of Casualties	Number of Vehicles	Road Surface Conditions	Speed limit	Time	Urban or Rural Area	Weather Conditions
-0.201349	1	2	Dry	30	15:11	Urban	Fine no high winds
-0.199248	11	2	Wet or damp	30	10:59	Urban	Fine no high winds
-0.179599	1	2	Dry	30	14:19	Urban	Fine no high winds
-0.20311	1	2	Frost or ice	30	08:10	Urban	Other
-0.173445	1	2	Dry	30	17:25	Urban	Fine no high winds
-0.185525	3	2	Dry	30	11:48	Urban	Fine no high winds

METHODOLOGY

Data Cleaning and Categorisation

- Removal of duplicate records
- Replacement of underscores with spaces using the *SUBSTITUTE* function
- Standardisation of date and year fields into numeric format
- Grouping of time, vehicle type, and lighting conditions into meaningful categories

Data Analysis

- Pivot Table
- Descriptive Analysis
- Defined KPI

Data Visualisation

- Use of charts, maps, and graphs to identify trends, patterns, and hotspots
- Development of an interactive dashboard for final visualisation and insight exploration

RESULTS

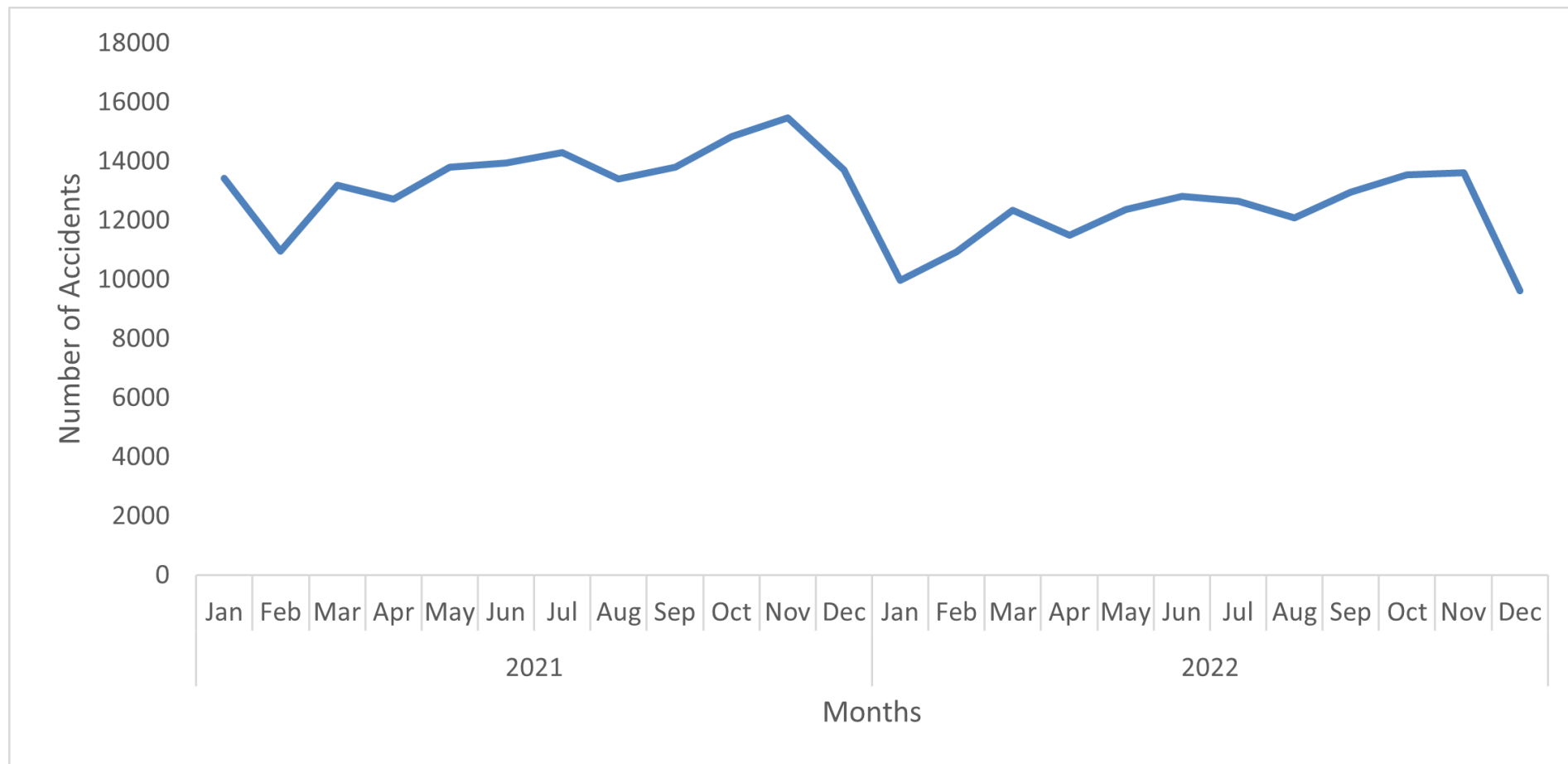


Figure 1: Monthly Road Accident Trends in the UK (Jan 2021 – Dec 2022)

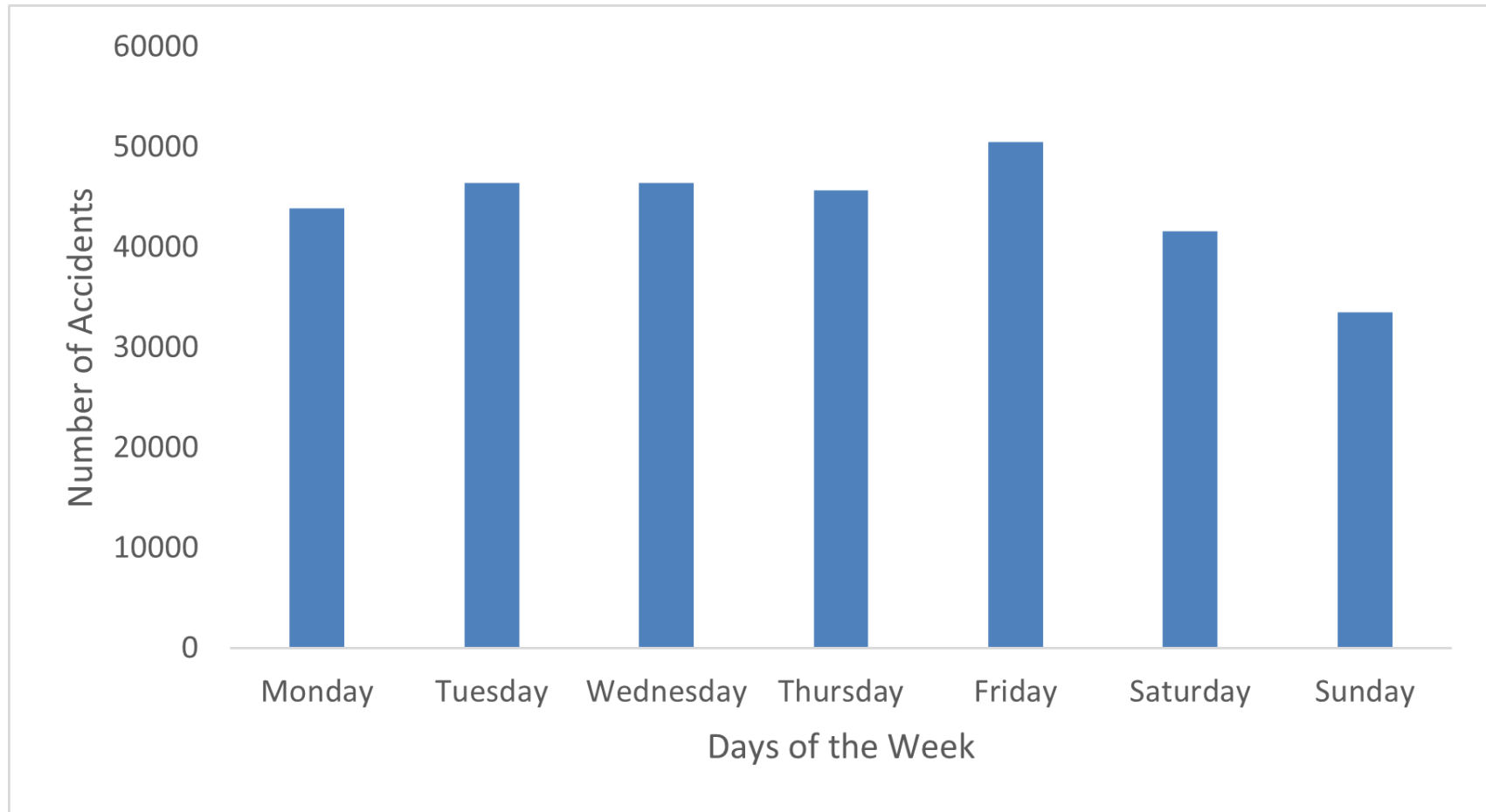


Figure 2: Total Number of Accidents by Day of the Week

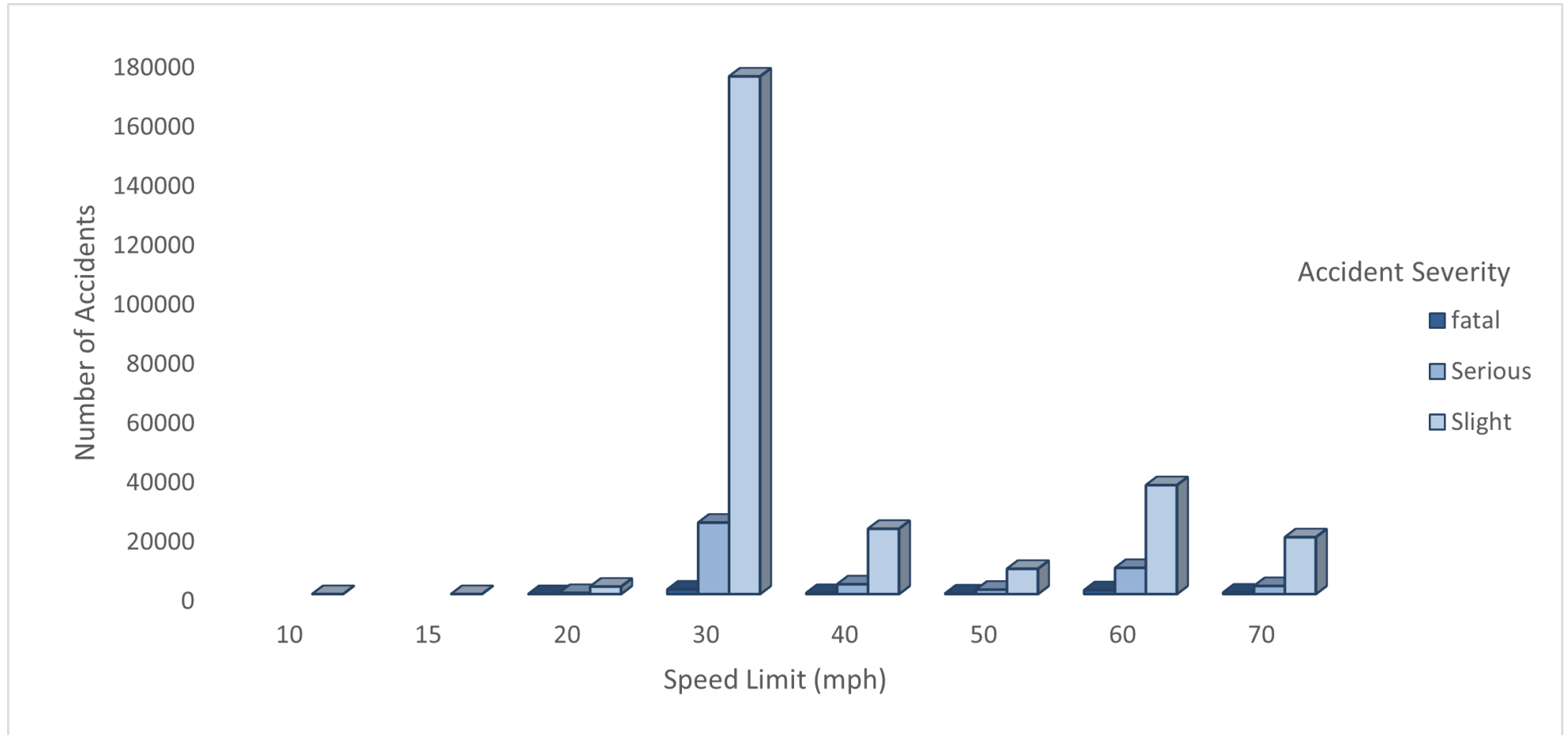


Figure 3: Total Number of Road Accidents by Speed Limit and Accident Severity

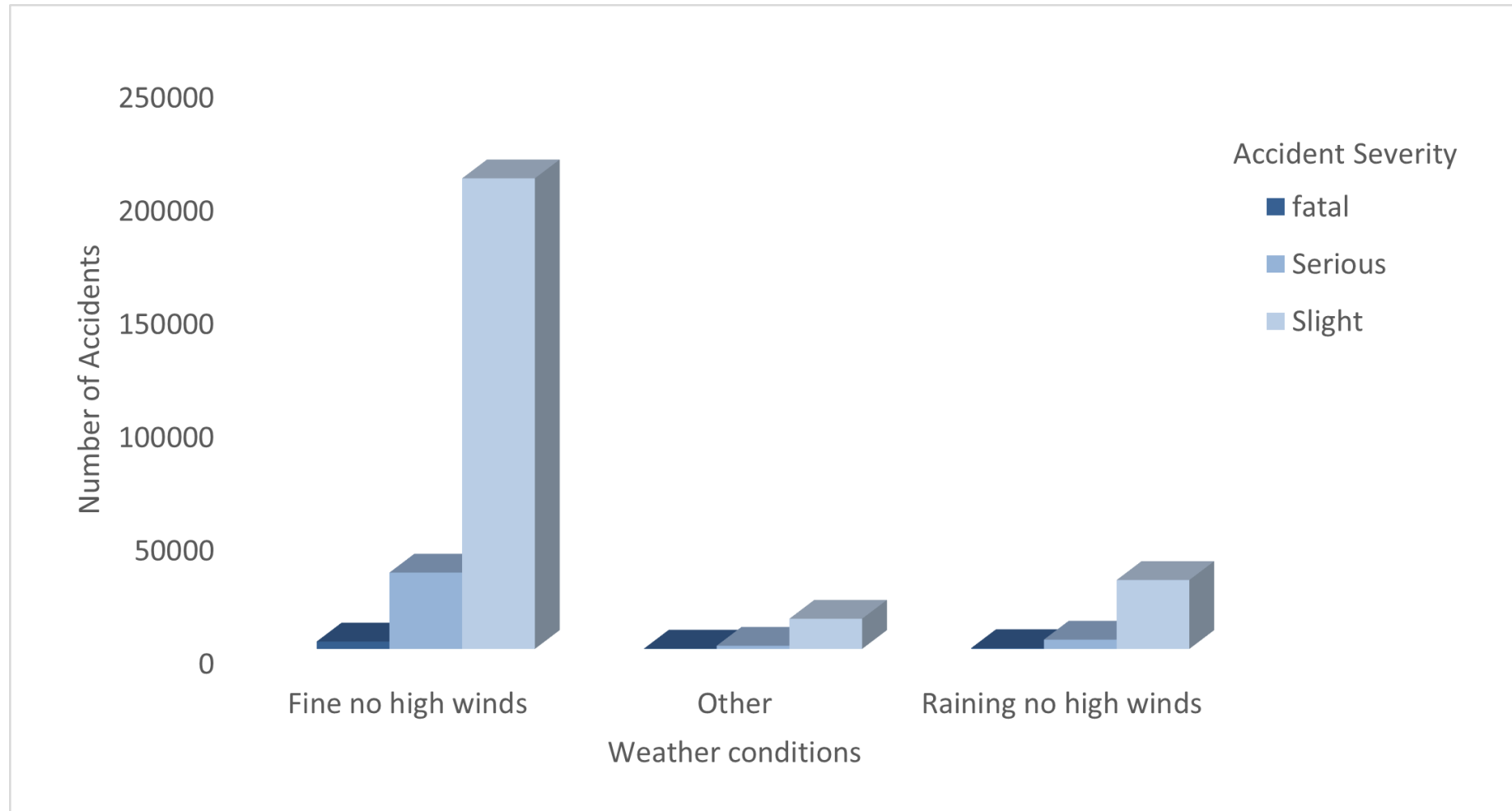


Figure 4: Total Number of Road Accidents by Weather conditions and Accident Severity

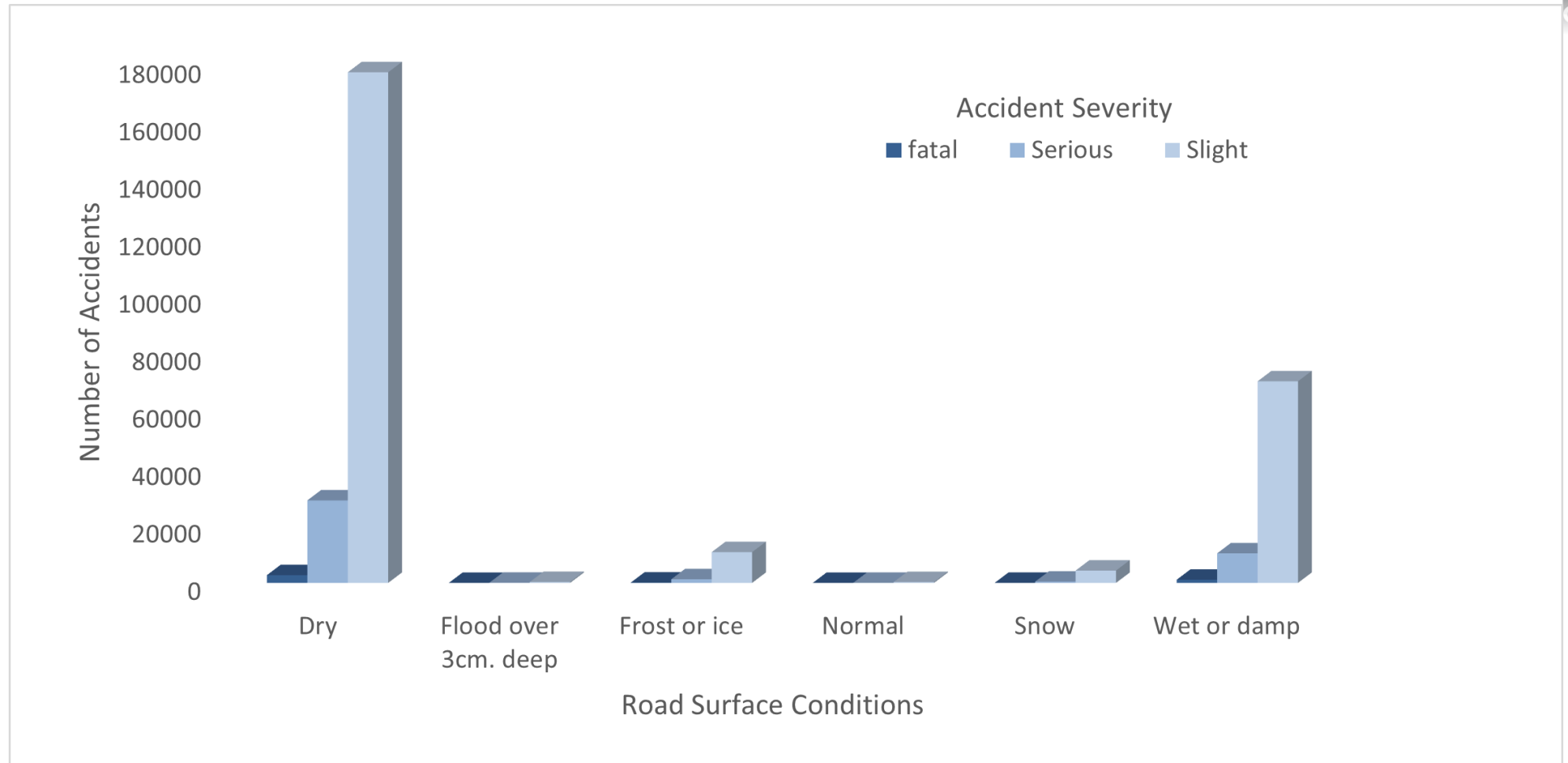


Figure 5: Total Number of Road Accidents by Road Surface Conditions and Accident Severity

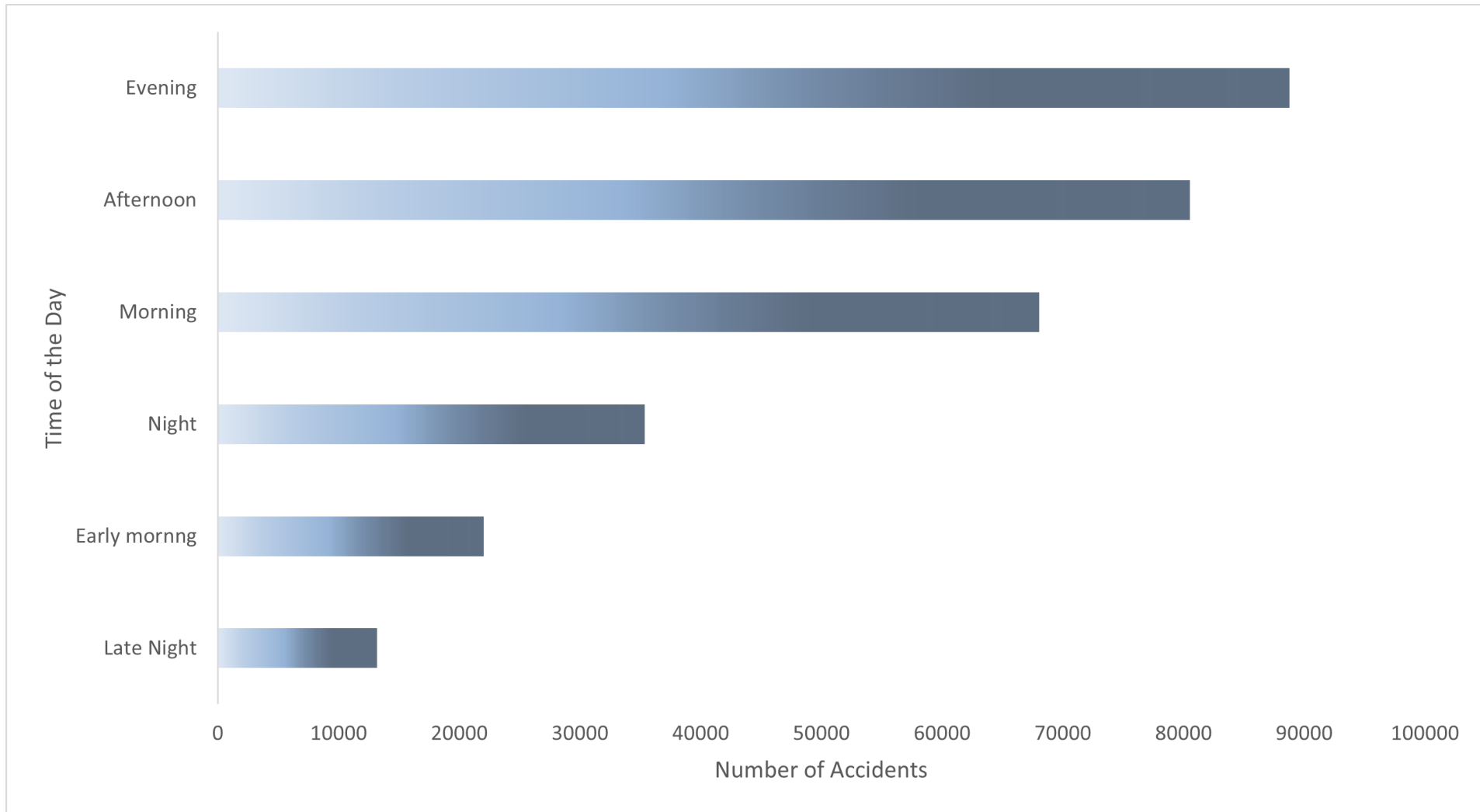


Figure 6: Total Number of Road Accidents by Time of the Day

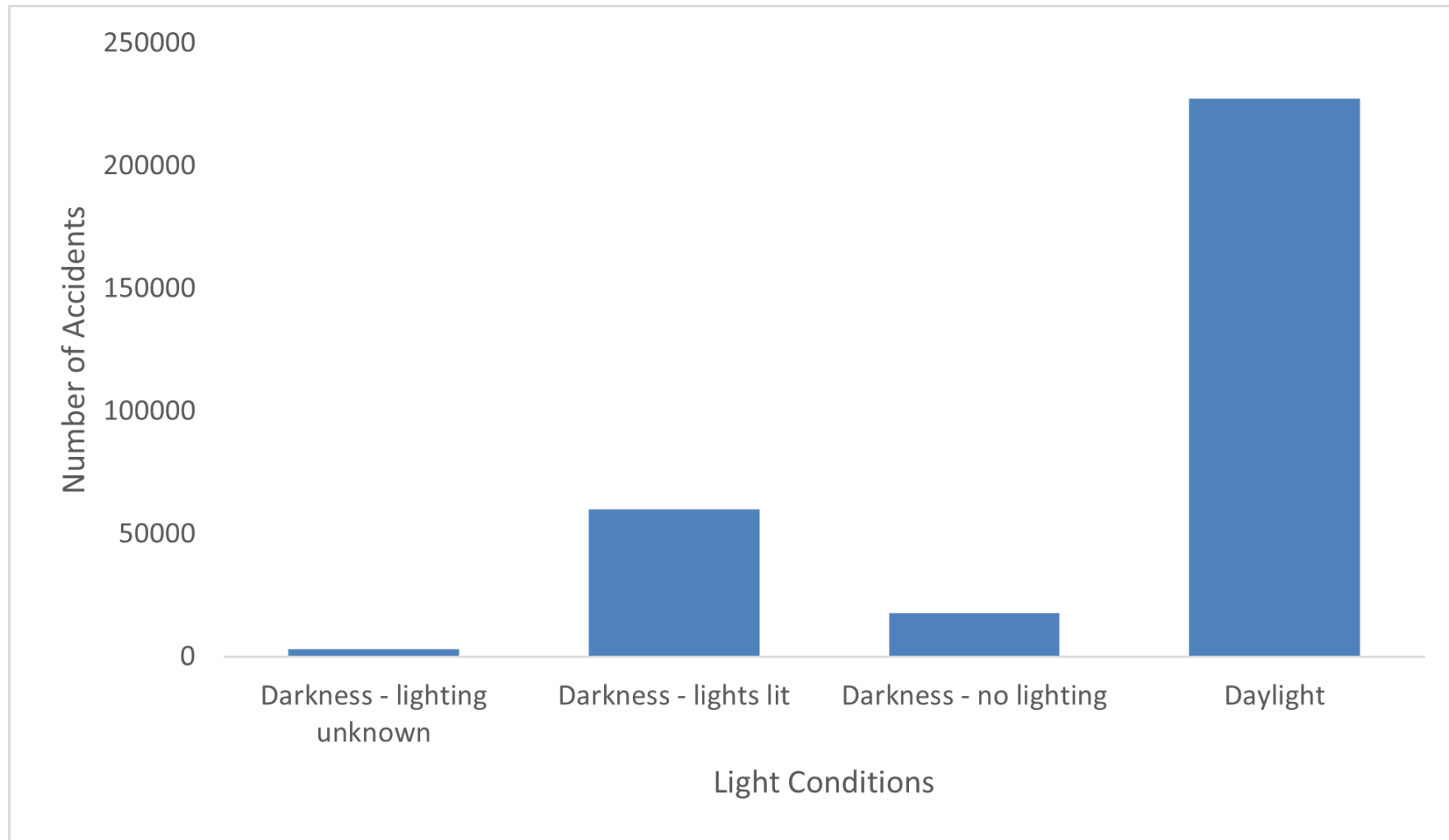


Figure 7: Total Number of Road Accidents during different Light Conditions

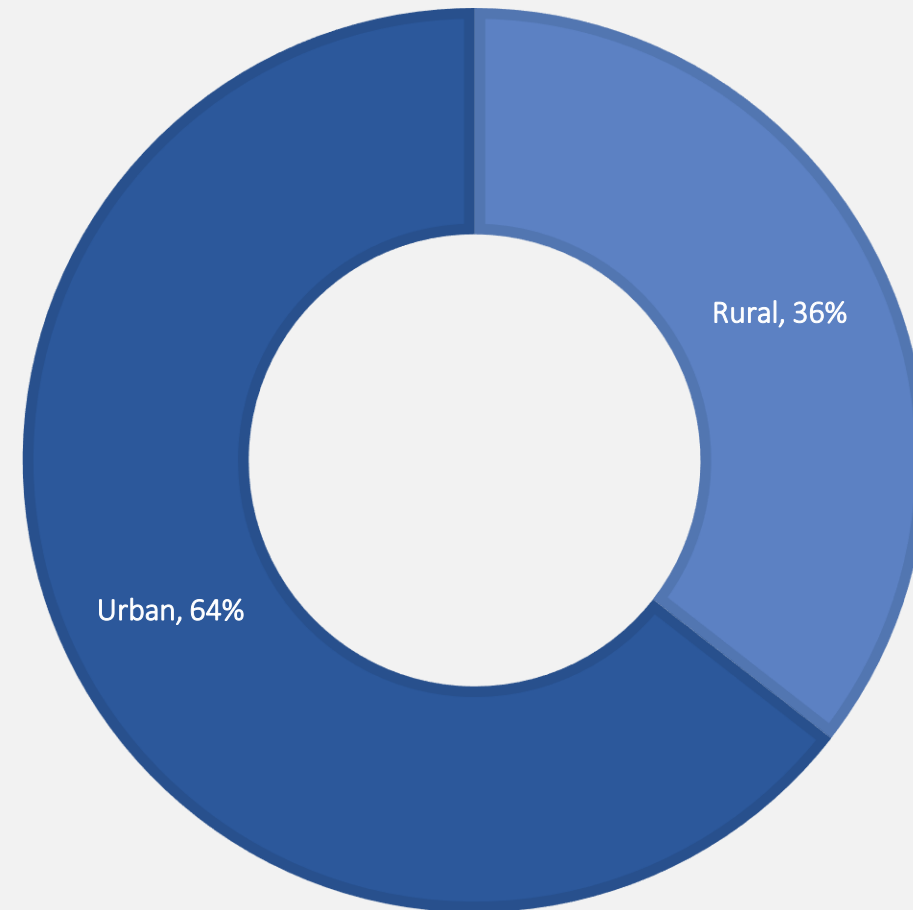


Figure 8: The Proportion of Road Accidents by Settlement Area

Descriptive Statistics	No. of Casualties	Speed Limit (mph)
Mean	1.36	38.87
Median	1	30
Standard deviation	0.82	14.03
Range	47	60
Skewness	5.69	1.14

Table 1: Descriptive Statistics: Casualties and Speed Limit

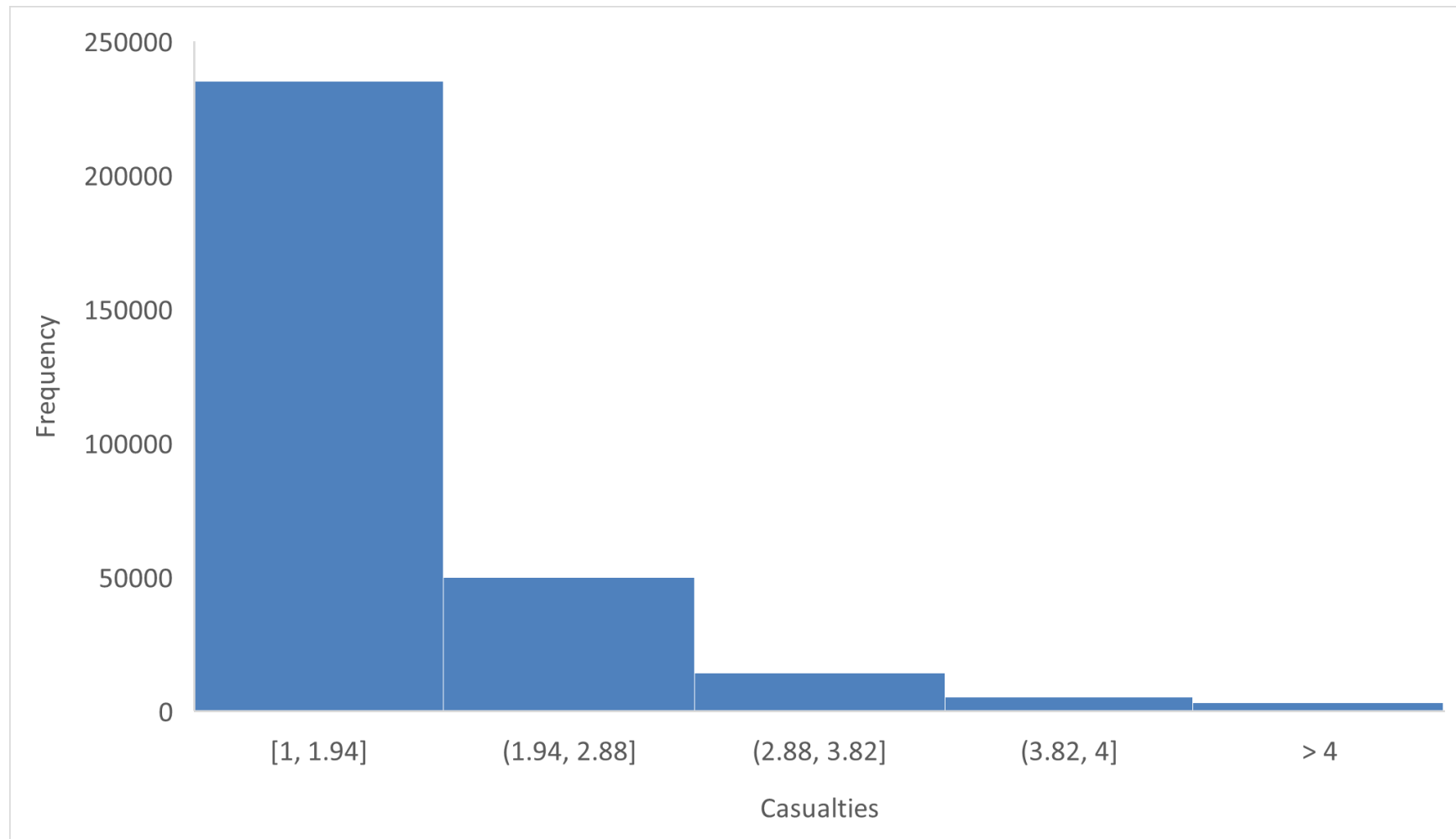


Figure 9: Casualties show a Right-Skewed Distribution with Kurtosis of 129.9 and Skewness of 5.7

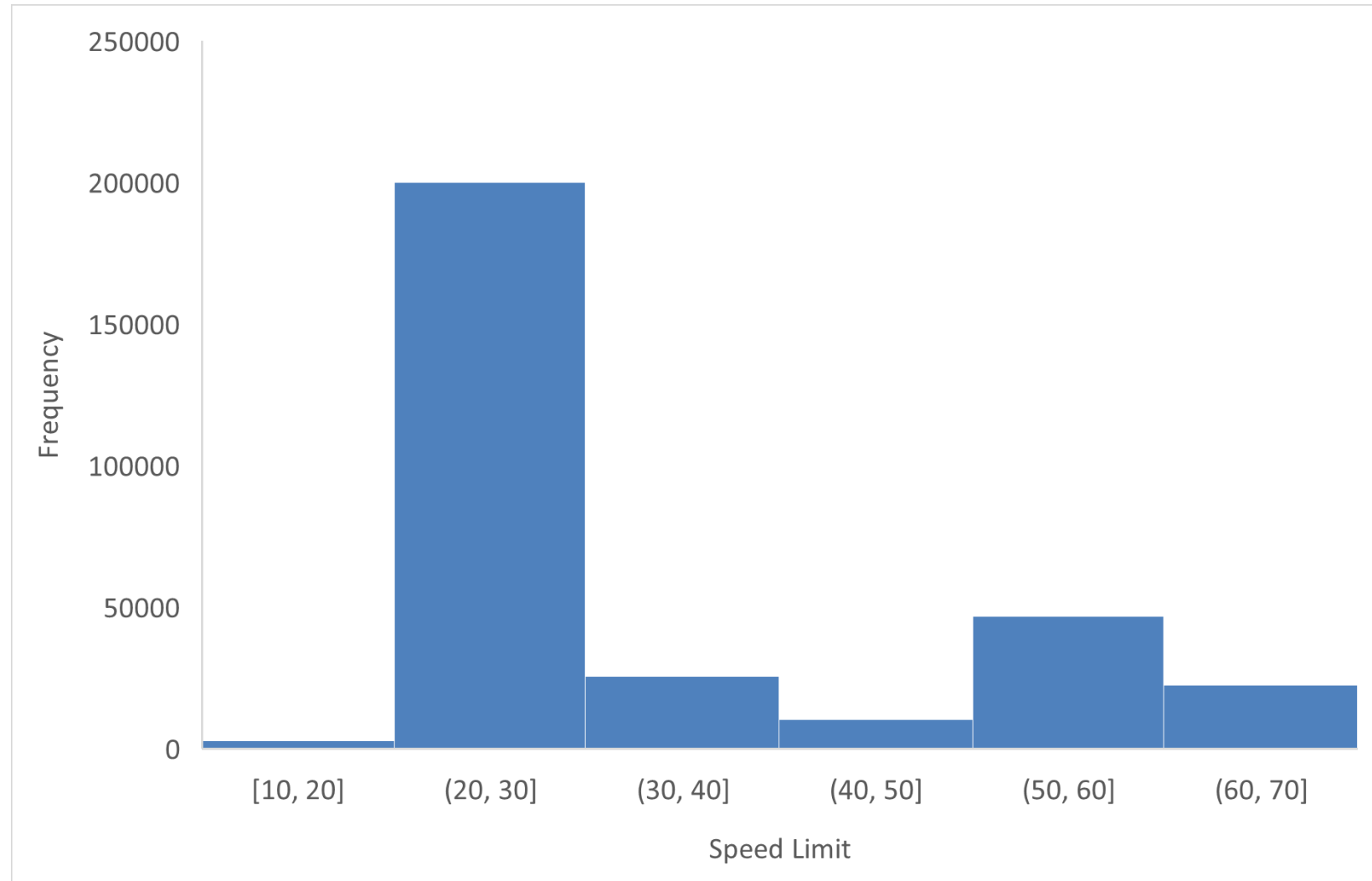


Figure 10: The Speed Limit shows a Right-Skewed Distribution with Kurtosis of -0.3 and Skewness of 1.1

KEY METRICS

**TOTAL
ACCIDENTS**
307972

**TOTAL
CASUALTIES**
417882

**FATAL
INJURIES**
3953

**SERIOUS
INJURIES**
40740

**SLIGHT
INJURIES**
263279

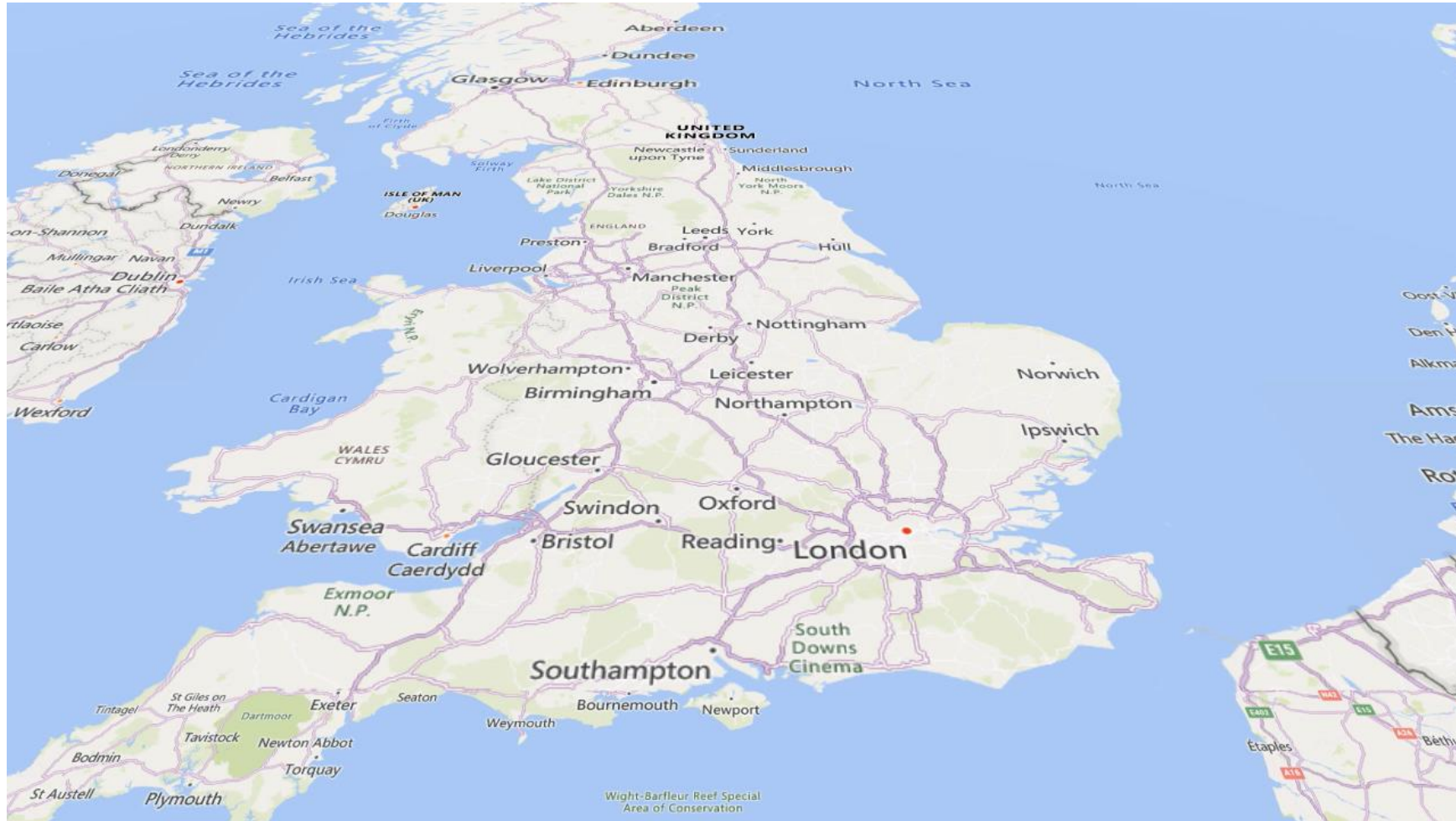


Figure 10: UK Road Accident Hotspots Heat Map (2021-2022)

DASHBOARD OVERVIEW

ROAD ACCIDENT DATA

Total Accidents
307972

Total Casualties
417882

Total Vehicles
563301

Fatal Injuries
3953

Serious Injuries
40740

Slight Injuries
263279

Cleaned Year

2022

Road Surface Conditions

Wet or damp

Speed limit

30

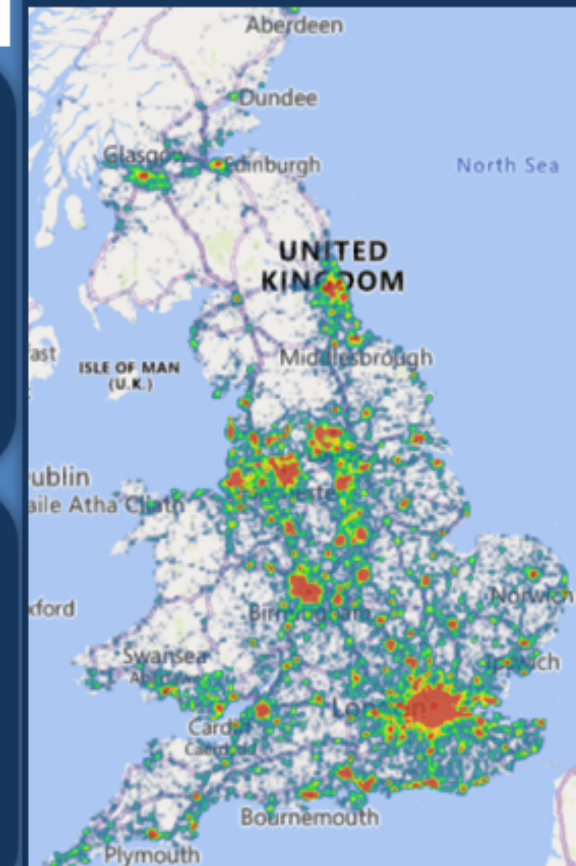
New Light Conditions

Daylight

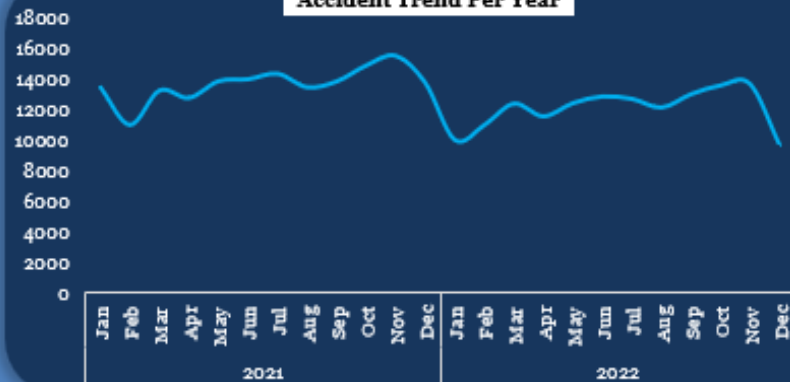
Standardized Time Band

Evening

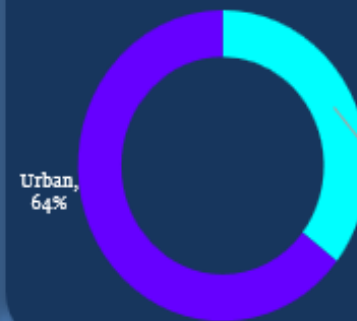
Accidents by Location



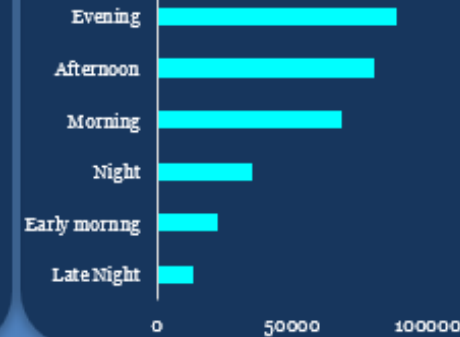
Accident Trend Per Year



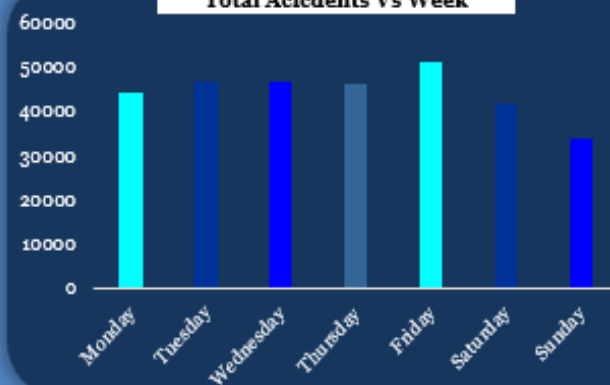
Urban vs Rural



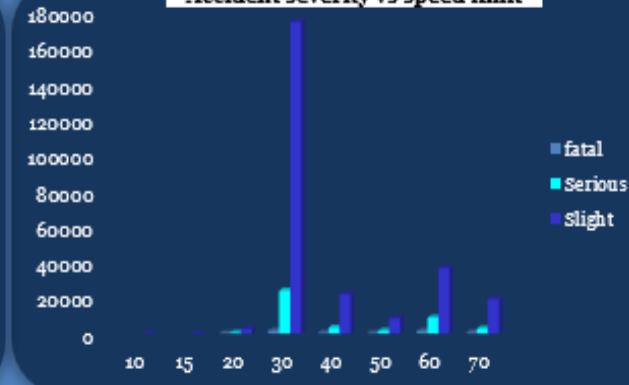
Total Accidents vs Time of Day



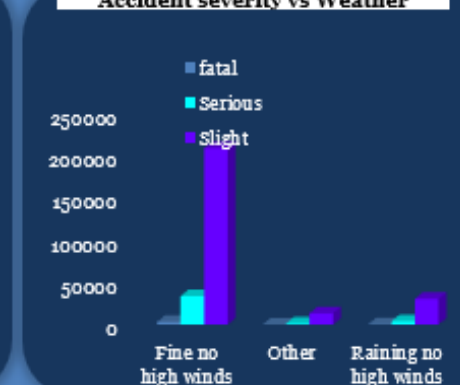
Total Accidents Vs Week



Accident severity vs speed limit



Accident severity vs Weather

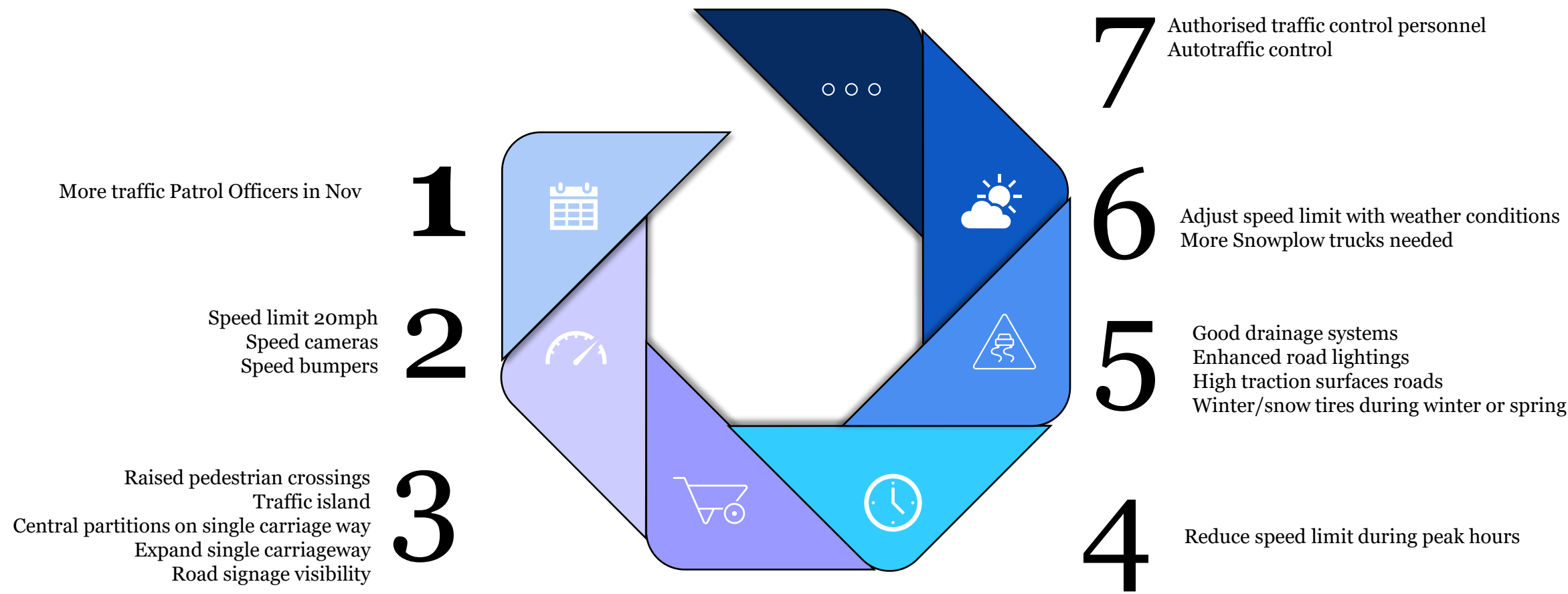


- The dashboard enables stakeholders to explore accident patterns interactively by time, severity and road conditions

DISCUSSION

- ❖ The analysis reveals relationships between environmental conditions, road features, and accident severity.
- ❖ Accident frequency peaks during afternoons, evenings, and Fridays, reflecting traffic flow patterns.
- ❖ Although the majority of accidents occurs on dry roads, adverse conditions such as rain, ice, wet roads, and poor lighting significantly increase accident risk.
- ❖ Higher speed roads (50 -70 mph) are more likely to be associated with serious and fatal injuries.
- ❖ Descriptive statistics indicate that most accidents involve a single casualty, while a small number of accidents result in very high casualty counts.
- ❖ Overall, the findings highlight the need for targeted interventions, improved infrastructure, and data-driven policy actions.

Road Safety Interventions for Accident Hotspots





Policy Measures



- Enforcement of speed management within 30 mph in urban and residential areas
- Penalties for repeat offenders
- Mandatory road safety audits for new and existing roads
- Promote evidence-based policy updates informed by accident data trends.

RECOMMENDATION



A

Policy Measures



- Enforcement of speed management within 30 mph in urban and residential areas
- Penalties for repeat offenders
- Mandatory road safety audits for new and existing roads
- Promote evidence-based policy updates informed by accident data trends.

B

Infrastructure Improvements



- Upgrade of uncontrolled junctions
- Roundabouts at high-risk junctions
- Improve junction visibility, road markings and signage for clearer driver guidance
- Improve night-time lighting
- Upgrade road surfacing in wet weather
- Pedestrian and cycling lanes
- Smart traffic signals

C

Public Awareness Strategies



Targeted public campaign:

- Awareness on dangers of speeding
- Wet-weather driving campaigns
- Educate drivers on night-time driving risk
- School-area safety programmes

Conclusion

- ❖ The analysis highlights clear patterns in accident frequency and severity across the UK
- ❖ Speed, environmental conditions, lighting, and road surface conditions emerged as key contributors to accident risk
- ❖ While most accidents result in minor injuries, a small proportion of high-impact accidents significantly increase overall casualties
- ❖ Applying data-driven interventions can support safer roads, reduce casualties, and improve decision-making
- ❖ This project demonstrates 10Alytics' ability to transform raw data into meaningful, actionable insights



TEAM POLARIS CLOSING REMARK

- ✓ **Kuro:** As a team, we used data to identify the key factors driving road accidents and the areas that need urgent attention.
- ✓ **Onos:** Our analysis revealed clear patterns in severity, timing, and environmental conditions that influence accident outcomes.
- ✓ **Tolu:** The findings show that targeted policies and stronger enforcement can reduce high-risk behaviours.
- ✓ **Dave:** We identified critical infrastructure gaps where improvements can significantly enhance road safety.
- ✓ **Ayo:** Public awareness and education remain essential to encouraging safer driving behaviour.
- ✓ **Ey:** This project highlights the value of data analytics in delivering meaningful road safety insights.

Pod Polaris



- Kuroyefa
Team Lead



- Onoriode
Deputy Team Lead



- Eyiche
Member



- Ayo
Member



- Dave
Member



- Tolu
Member

THANK YOU